OSSP PRACTICAL PROGRAM2: -

Develop a C program that uses the system calls open(), read(), write(), and close() to copy the contents of one file to another. Explain how control transitions between user space and kernel space during execution.

Use the strace utility to trace all system calls generated by the command cat sample.txt. Analyze the sequence of system calls and identify the kernel services involved.

Code: -

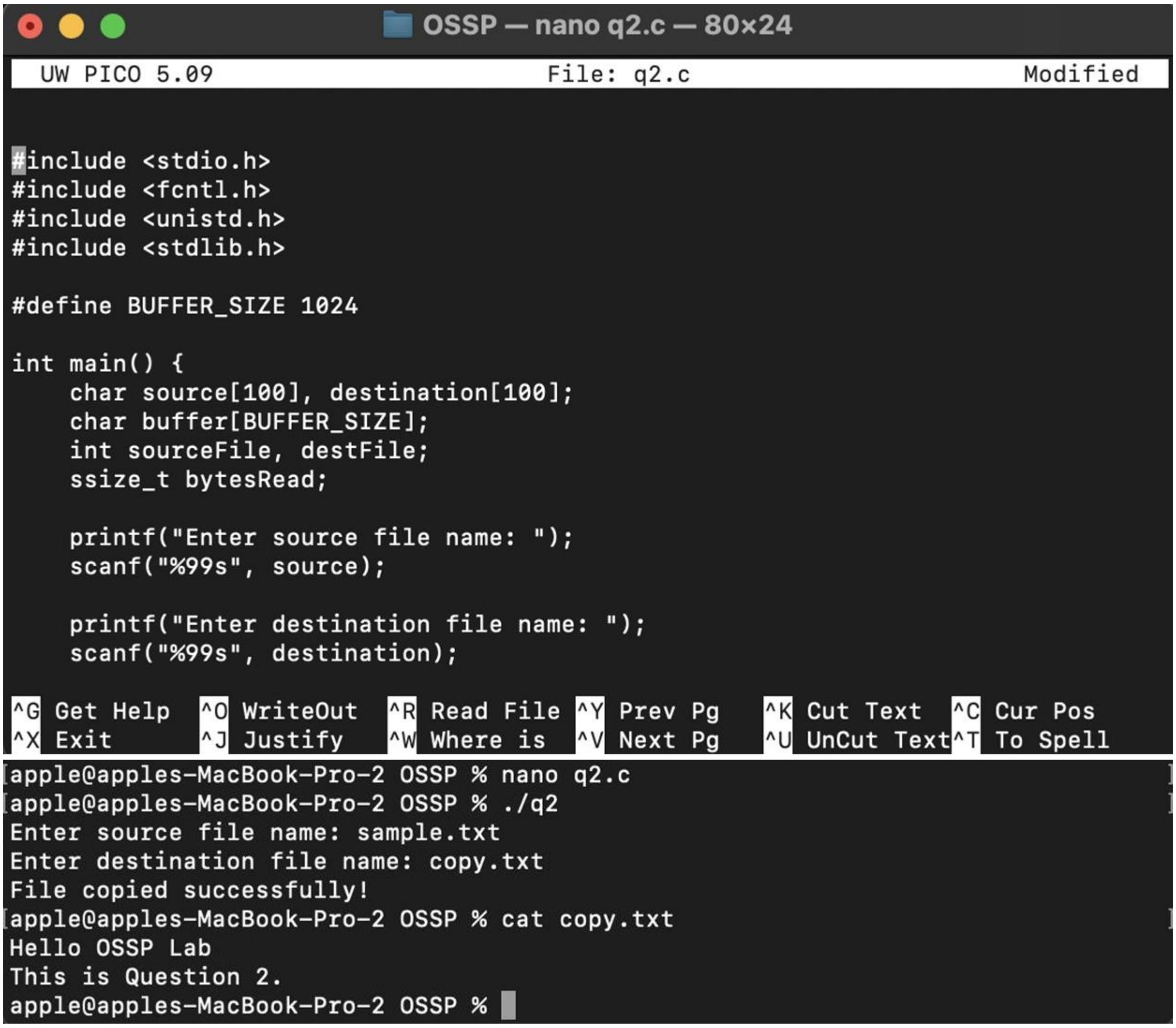
```
#inclue<stdio

.h>


#include

<stdlib.h>

#include

<fcntl.h>

#include

<unistd.h>




int main() {


  int source,

  destination; char

  buffer[1024];
```

```
ssize_t

bytesRead;

source = open("source.txt", O_RDONLY);
if (source == -1) {
  perror("Error opening source
  file");

  return 1;

}
destination = open("destination.txt",
          O_WRONLY | O_CREAT |
          O_TRUNC, 0644);

if (destination == -1) {

  perror("Error opening
  destination file");
  close(source);

  return 1;

}

while ((bytesRead = read(source, buffer,
  sizeof(buffer))) > 0) { if (write(destination, buffer,
  bytesRead) != bytesRead) {
```

```
        perror("Error writing to
        destination file"); close(source);
        close(destination);
        return 1;
    }
}
if (bytesRead == -1) {
    perror("Error reading source file");
} else {
    printf("File copied successfully!\n");
}
close(source);
close(desti
nation);
return 0;
```

}

OUTPUT: -


OSSP — nano q2.c — 80×24
UW PICO 5.09
File: q2.c
Modified
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>
#include <stdlib.h>
#define BUFFER_SIZE 1024
int main() {
char source[100], destination[100];
char buffer[BUFFER_SIZE];
int sourceFile, destFile;
ssize_t bytesRead;
printf("Enter source file name: ");
scanf("%99s", source);
printf("Enter destination file name: ");
scanf("%99s", destination);
^G Get Help
^O WriteOut
^R Read File
^Y Prev Pg
^K Cut Text
^C Cur Pos
^X Exit
^J Justify
^W Where is
^V Next Pg
^U UnCut Text
^T To Spell
apple@apples-MacBook-Pro-2 OSSP % nano q2.c
apple@apples-MacBook-Pro-2 OSSP % ./q2
Enter source file name: sample.txt
Enter destination file name: copy.txt
File copied successfully!
apple@apples-MacBook-Pro-2 OSSP % cat copy.txt
Hello OSSP Lab
This is Question 2.
apple@apples-MacBook-Pro-2 OSSP %